

## Education

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**Hamilton, ON** **McMaster University** **Sept 2015 - Present**

- Bachelor of Engineering: Electrical Engineering (Co-op) - GPA 3.7
- Coursework: Data Structures, Algorithms, Principles of Programming, Engineering Computation, Logic Design, Calculus 3, Engineering Profession & Practice, Engineering Design & Graphics

## Employment

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**Robotics & CS Counselor** **City of Markham** **Summer 2016**

*Markham, ON*

- Utilized Scratch & Ozoblockly to introduce essential programming concepts to students
- Introduced the concept of an algorithmic approach to solve difficult problems
- Developed excellent communication & leadership skills, through teaching content to a young audience

**Camp Counselor** **Camp Green Acres** **Summer 2015**

*Markham, ON*

- Maintained accurate records of events including: attendance, incident reports, & logbook documents
- Led daytime activities within the entire camp, & followed a detailed schedule in a strict manner
- Communicated with parents about participants' experiences & raised concerns to members of staff

**VP Training- CS** **Pierre Elliott Trudeau HS** **Fall 2014 - Spring 2015**

*Markham, ON*

- Taught core computer science topics encompassing: object oriented programming, complex data structures, algorithms, time complexity, & Java as a core programming language
- Held classes explaining the principles of computer programming to students without prior knowledge
- Led a team of 4 students, while simultaneously designing lesson plans & coding workshops

## Projects

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- **Medieval War Simulation:** Implemented a path-tracking algorithm for an AI to imitate player behavior, applied principles of objected oriented programming, & utilized Java in the Greenfoot IDE. <https://github.com/hansenngai/Medieval-War-Simulation>
- **Android Interview App:** An application that allows filtering through user information from database indices of profiles, stored through data entry into a JavaScript program. Utilized Android Studio & accessed parsed JSON data to update the app continuously
- **CD Rom Gear System:** Designed to fit inside a regular chassis, followed strict physical constraints, & completed a thorough engineering report. Innovated the system through usage of Autodesk Inventor & MapleSim as part of product development process

## Activities

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- Formula Electric Racecar- Low Voltage Team
- MLH (Major League Hacking) Participant
- McMaster Engineering Society
- Student Mentorship Program (Under PEO)

## Languages & Technologies

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- Java, Python, C, HTML, CSS, JavaScript, MATLAB, Android

